

Sensors, Data Collection, and Synchronization

Section 2 of the presentation

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Why We Added Sensors

- A frontal 2D camera is useful, but it cannot give reliable ground truth for knee orientation.
- So we added two BWT901CL IMU sensors, one near each knee.
- The goal is to collect synchronized video and IMU data during the drop jump.
- These sensor signals become the reference for future supervised learning.

Key idea

Video estimates movement. IMU provides a reference for knee pitch, roll, and yaw.

Sensor setup

Two BWT901CL sensors placed on the knees for drop-jump recording.

Data Collection Pipeline



- The pipeline follows the same logic as the main jump-analysis workflow.
- Recording starts automatically only when the real drop is detected.
- If setup or protocol checks fail, the trial is rejected and not saved.

What We Save for Each Valid Trial

movement_timeseries.csv

Frame-by-frame pose trajectory, video-derived knee orientation proxies, aligned left/right IMU pitch-roll-yaw, and body-scale values over time.

front_2d_features.csv

The 37 frontal features used by the vision pipeline, extracted only from valid trials.

trial_metadata.json

Setup calibration, protocol checks, participant settings, and sensor acquisition status.

Raw IMU

When live, right raw data is also stored.

- Each trial folder combines vision data and sensor data in one structured package.
- This keeps collection separate from final model inference.

Synchronization and Why It Matters

- Video and IMUs can have different sampling rates.
 - The first video frame is used as the jump start reference.
 - IMU timestamps are converted to time-from-start.
 - Sensor values are interpolated onto video timestamps.
 - Each video frame gets matched sensor measurements.
- The dataset is cleaner because in
 - The IMUs provide ground truth for
 - Later models can correct or replace orientation estimates.
 - So this part builds reliable training